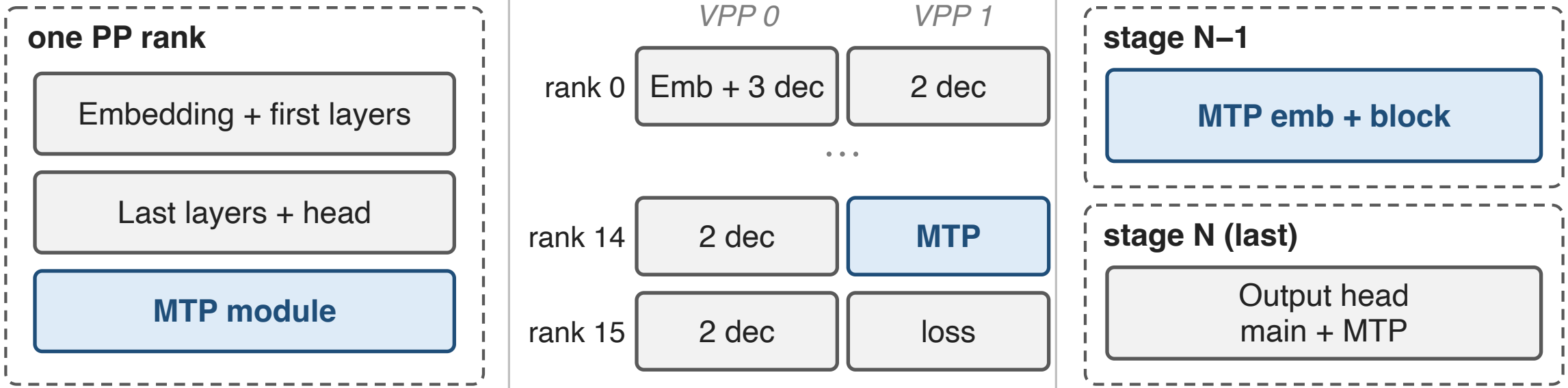
one PP rank
Embedding + first layers
Last layers + head
MTP module
(a) DeepSeek-V3: DualPipe
VPP 0
VPP 1
rank 0
Emb + 3 dec
2 dec
rank 14
2 dec
MTP
rank 15
2 dec
loss
(b) MCore: own virtual stage
stage N–1
MTP emb + block
stage N (last)
Output head
main + MTP
(c) GLM-5: split MTP